

RESEARCH BRIEF

8 Trends Shaping The Future Of Blockchain Technology

May 8, 2018

[Bitcoin & Blockchain](#)



Regulatory scrutiny is increasing, but this isn't slowing the pace of blockchain activity — if anything, it's helping the industry mature. As financing methods shift and investors search for the killer app, we dive into eight trends shaping the future of blockchain technology.

WHERE IS THIS DATA COMING FROM?

Start your free trial today

SIGN UP

While the prices of cryptocurrencies and cryptoassets have fallen from their 2017 peaks, equity investment into blockchain startups is on pace to hit all-time highs in 2018.

Even as the media storm around the sector has calmed, regulators are acting more aggressively and decisively, issuing subpoenas to bad actors and providing guidance to good ones. And, although Google searches for “bitcoin” and “blockchain” have slid, companies are still flush with cash from 2017’s ICO boom, cryptocurrency price run-up, and venture capital investment — and are spending it.

THE FUTURE OF BLOCKCHAIN TECH

See how venture firms, corporates, regulators, and builders are shaping the future of blockchain technology.

DOWNLOAD NOW

In this report, we'll dive into eight key trends shaping the future of blockchain.

Specifically, we'll talk about:

- **Initial coin offerings**
 - How venture firms are increasingly investing in once-taboo tokens and ICOs
 - Regulatory activity, and how it's a double-edged sword for industry players
- **Venture activity**
 - Where top VCs are placing their blockchain bets — from infrastructure to cats
 - The rise of security tokens as a distinct asset class
 - How 2017's winners are turning into money managers, venture investors, and acquirers
- **Corporate activity**
 - Which big corporates might be bluffing on earnings calls
 - Investment growth and a bad case of déjà vu
 - Which consortia are forging ahead, despite challenges

Initial coin offerings

Initial coin offerings (ICOs) are sales of tokens by blockchain companies looking to raise funds. According to teams holding ICOs, tokens provide access to, or utility within, a decentralized ecosystem. Tokens are scarce, and could rise in value if demand outweighs supply.

As an example, if the network in question is – say – a decentralized social network, tokens might grant access. If more people want access, then the token's value might rise as people buy and sell their tokens.

Given this context, let's dive into some key trends.

Where VCs trade cash for equity, ICO investors trade cash for tokens. Although regulators grapple with this definition, we'll use it for our purposes to understand how this financing method is changing.

Increasingly, the line between equity financing and ICOs is blurring. Traditional equity investors are investing in companies that plan on using tokens within their business models. Further, traditional investors are acquiring tokens outright via pre-sales, SAFT contracts (described below), and regulatory-compliant offerings – something that seemed out of the question months ago.

Pre-sales are rounds held before larger, public ICOs. These don't follow a uniform structure. In some cases, pre-sales offer discounted tokens to early investors (accredited and unaccredited). In others, teams sell small equity stakes in exchange for runway before an ICO. ICOs are expensive, and often incur legal, marketing, and advisory expenses. Venture funded pre-sales could cover expenses before an ICO infuses cash into the company.

In still other cases, accredited investors buy tokens via cryptocurrency purchase agreements. A common version of this is the SAFT (Safe Agreement for Future Tokens). The SAFT acts as a forward contract for tokens, with the contract converting upon deployment of a functioning network. Thus, startups aren't selling equity stakes, but the rights to some of the tokens it will use as part of its network. Such a conversion might happen years after the initial sale.

The rise of pre-sales and SAFTs reflects venture funds wanting to cash in on the token economy. Recall that the median time between first funding and IPO for VC-backed tech companies that went public in 2017 was about 9 years. Tokens trade on exchanges (often before the network launches), and provide near-immediate liquidity. As for the SAFT, a token conversion could happen within one or two years, also providing quick liquidity to venture investors.

Additionally, many venture firms have mandates that require them to invest in specific asset classes – namely, private companies. SAFTs and certain pre-sales might be a way to invest in tokens while still conforming to this asset class requirement, whereas buying tokens via an exchange or directly via an ICO likely would not meet those requirements.

A shift toward pre-sales is borne in the data, with venture rounds trending up and pure-play ICOs trending down.

In February, ICOs raised almost 60% of their capital in private rounds and pre-sales, according to TokenData. Some of that data is captured below, with "other" VC equity deals up considerably as a share of blockchain deals. These include some private rounds and pre-sales.

Lots of recent equity dollars go to pre sales, and for runway before large, public ICOs

Equity financing (excluding ICOs). January 2017 – March 2018

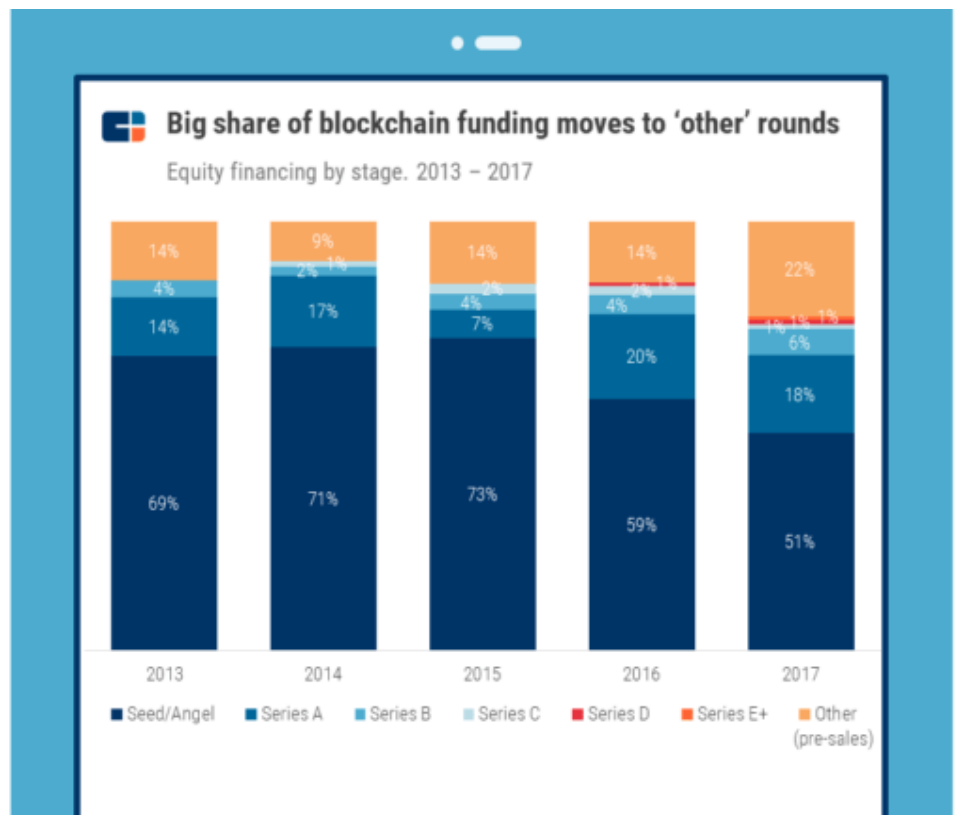


Pre-sale financings are on the rise

Lines are blurring between equity and ICO financing, as venture investors increasingly deploy capital to companies in pre-sales and private rounds.

In short – the token model is emerging as a de facto fundraising mechanism for blockchain teams.

CBINSIGHTS



For more evidence, pure-play ICO deals and dollars are down; in March, 113 ICOs closed for about \$500M, down from 121 and \$1.2B in February. Put another way, ICOs that closed in February raised an average of about \$10M, while in March they raised an average of \$4.5M. As a caveat, these numbers are somewhat lagging indicators – we only count completed ICOs, and not those that are still ongoing.

Future play ICOs are losing funding momentum

Disclosed funding of completed ICOs. January 2017 – March 2018



CBINSIGHTS

Sources: TokenData.

22

In one notable example of the trend toward pre-sales, Telegram held such a large pre-sale, that it no longer has plans for a public sale. The encrypted messaging service raised \$1.7B from 175 private investors in two separate private rounds, selling “purchase agreements for cryptocurrency” (not equity).

Another blockchain team, Basis (fka Basecoin), raised \$125M from 225 investors via a SAFT sale that took place starting at the end of Q1’18. Basis is building a “stablecoin” that is intended to be less volatile than other cryptocurrencies.

With all this, pre-sales are a mixed bag. On the one hand, they’re still relatively risky, and regulators still haven’t fully weighed in on them. On the other, they shift risk onto accredited and venture investors, a good thing for consumers and regulators. As regulators continue to crack down on public ICOs, we expect this shift toward pre-sales and private sales to continue.

REGULATORY ACTIVITY IS A DOUBLE-EDGED SWORD — BAD FOR ICOS, GOOD FOR OTHERS

ICO evangelists like to argue that ICOs are new financial instruments which require new legislation and regulatory agencies. So far it looks like US regulators do not agree. US regulatory agencies, including the SEC, CFTC, and FinCEN, have generally reacted skeptically and unfavorably toward the new financing mechanism. At the same time, regulators’ increasing scrutiny of the sector is actually encouraging new companies to enter the sector.

The SEC, CFTC, and FinCEN (in addition to other government agencies) seem to be approaching the sector in different ways.

of hallmarks of a security.” Three months later, in February, Mr. Clayton said before Congress: “I want to go back to separating ICOs and cryptocurrencies. ICOs that are securities offerings, we should regulate them like we regulate securities offerings. End of story.” At around the same time, the SEC subpoenaed a number of cryptocurrency hedge funds and organizations that held ICOs.

From its rhetoric and activity, it appears that the SEC is treating ICOs as securities. This doesn't bode well for teams that held them, or their legal advisors. These companies could now face legal repercussions. At the same time, it appears that the SEC is treating some existing cryptoassets as currencies or commodities — not securities.

ICO MARKET IS LIKELY IN FOR A RUDE AWAKENING

“I have yet to see an ICO that doesn't have a sufficient number of hallmarks of a security.”



Jay Clayton
Chairman, SEC
November 8, 2017

CBINSIGHTS

FinCEN, the Financial Crimes Enforcement Network, sent a letter to Senator Ron Wyden (who sits on the Senate Committee on Finance) in March, writing that “a developer that sells convertible virtual currency, including in the form of ICO coins or tokens, in exchange for another type of value that substitutes for currency is a money transmitter and must comply.” Such an opinion — of cryptocurrencies as money — contrasts with the CFTC's definition, of cryptocurrencies as commodities. FinCEN's conclusion puts teams issuing tokens at significant risk for potential fines or other repercussions.

Above all, these three different definitions highlight regulatory disagreement and confusion. Industry policy organizations (like Coin Center) have been quick to offer educational resources around blockchain technology to legislators and regulators. Prominent VCs Andreessen Horowitz and Union Square Ventures even privately proposed a token sale “safe harbor” to the SEC.

and securities is contributing to two developments.

First, the SEC's increased regulatory scrutiny has led many crypto companies to take a more cautious approach to financing. Legitimate crypto companies are moving back toward venture funding and avoiding pure-play ICOs, while others are pursuing ICOs but making sure their token launches are compliant. One platform providing compliance solutions is **Templum**, which raised nearly \$13M in two rounds. Templum provides "regulatory compliant solutions for Tokenized Asset Offerings (ICOs as securities) and subsequent secondary trading."

Other attempts at self-regulation are found in SEC filings; companies are filing them more often. In one analysis, 12 "crypto" and "blockchain" related companies filed documentation with the SEC in 2017, and 23 have already filed in 2018.

A second effect of the SEC's stance — and broader regulatory clarity — is it's emboldened some established companies to make first moves in the sector. Square's Cash app now allows users to trade bitcoin, while Robinhood offers trading for a number of cryptoassets. In early May, Goldman Sachs announced that it's opening a bitcoin trading operation.

Ultimately, regulation appears to be a bust for unregulated public ICOs, and a boon for most others. Expect regulators to crack down with increasing regularity on ICOs, with both fines and indictments. At the same time, industry attempts at self-regulation are encouraging. Expect industry actors and regulatory agencies to converge on better-defined legislation and regulation.

Venture activity

Venture activity — equity financing, not ICOs — to blockchain companies hit its highest point in 2017, with 230+ deals closing for over \$1B. In total, 141 pure-play venture firms invested, alongside 119 corporations and their venture arms.

Below, we dive into three trends affecting venture investment to the sector.

TOP VCS ARE EMBRACING TOKENS, MOVING FROM "BLOCKCHAIN" TO "CRYPTO"

The rise of ICOs has pushed VCs to reconsider their traditional financing mechanism. As Fred Wilson (a prominent partner at Union Square Ventures) wrote on his blog, "With a new model, tokens, in its infancy, it begs the question of how it will impact our approach."

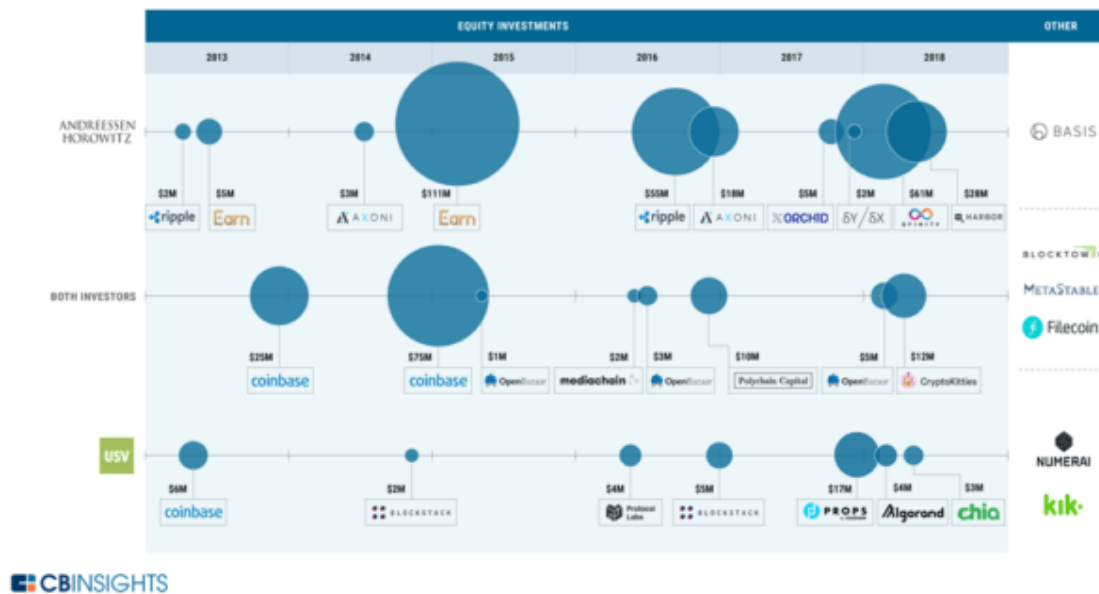
Indeed, venture firms are increasingly investing in tokens. As evidence, two of the sector's most active VCs — Andreessen Horowitz and USV — are increasingly investing in public blockchains and cryptoassets broadly.

Both firms have invested in cryptocurrency hedge funds as equity investors (Polychain Capital) and as limited partners (MetaStable). Andreessen is also reportedly preparing to launch a dedicated

Below, we show how their approaches to the sector have evolved. We highlight equity investments, investments into cryptocurrency hedge funds, and pivoting portfolio companies. Both firms' recent investments reflect a shift toward "crypto" as opposed to "blockchain technology." Some of these bets — such as Andreessen's bet on Basis — used SAFT contracts.

'Smart money' VCs are betting on tokens

Andreessen and USV's blockchain equity investments. 2013 – 2018 YTD (04/11/2018)



Looking at USV's bets, Fred Wilson explored his thesis in a recent blog post: "If the second half of 2016 and all of 2017 was about raising capital to fund development efforts (and speculating on all of that), then it sure feels like 2018 is the year we start getting decentralized applications we can use."

Indeed, some of USV's investments in the sector have to do with decentralized applications (dApps). **CryptoKitties**, for example, is a dApp that issues crypto-collectible digital cats. Its \$12M Series A at the end of March also saw participation from Andreessen. Another example is **OpenBazaar**, which is building a decentralized marketplace. OpenBazaar has received repeat capital from both investors.

USV has also bet on two cryptocurrency teams that hope to compete with Bitcoin. **Algorand** raised a \$4M seed round in February to build a high-speed payments protocol, as Bitcoin's faced transaction delays. **Chia** is a new "green" cryptocurrency that uses a more energy-efficient consensus mechanism than Bitcoin's "Proof of Work." The company raised a \$3.4M seed round at the end of March.

Turning to Andreessen Horowitz, the firm has more definitively shifted its strategy in the sector over time.

Andreessen's early bets focused on bitcoin and cryptocurrency trading. The firm participated in multiple rounds to **Earn** (fka 21), which was first a bitcoin mining company, and popular exchange

in contrast, Andreessen's recent blockchain gambles are infrastructure plays. **Dfinity** hopes to be an Ethereum competitor, and **Orchid** is, effectively, working on a private, decentralized internet. Harbor is deploying a protocol for "security tokens" (more below).

Of note, Dfinity held a "seed stage" ICO in February 2017, only later raising \$61M from Polychain Capital and Andreessen Horowitz. According to its CEO Dominic Williams, the company is wary of running another public ICO, given regulatory activity.

In sum, Andreessen's strategy has seen a defined shift — from Bitcoin, to blockchain, to crypto. USV's thesis has shifted less, with the firm never investing in enterprise blockchain providers. Both firms' shift toward tokens — first via cryptocurrency hedge funds, and now via more direct investments — is notable, and firms like Sequoia Capital, Bain Capital Ventures, and Founders Fund are also investing. Given the class of names getting involved, this trend will likely continue.

SECURITY TOKENS ARE SEEING INCREASED INVESTOR INTEREST

Regulations have also given rise to a new class of companies building security tokens platforms.

Security tokens are exactly what they sound like; securities on a blockchain. A security token could digitally represent any number of real-world assets, from real estate or vehicular title, to shares of a company. These are markedly different than utility tokens, which represent access to, or utility within, a given network. Most importantly, security tokens are subject to securities regulations.

Security tokens are also "programmable." This means that code can determine how they're used. For example, a loan — represented digitally — might automatically pay out according to an agreed-upon repayment schedule. Such code might execute without the use of a traditional middleman, like a bank.

Another advantage is liquidity. Tokenizing illiquid assets — like a venture fund, or a car — can make it easier to trade these assets over the web. Easier trading reduces friction, which correlates to increased liquidity.

One company working on this is Harbor. Harbor's \$28M Series B in April saw participation from a who's who of the VC world, including Andreessen Horowitz and Founders Fund. Harbor's "R-Token Standard" encodes rules in tokens. These only allow eligible investors to invest, and require them to follow KYC/AML regulations, among others.

Another company operating in this sphere is Polymath. Polymath hopes to help "trillions of dollars of financial securities migrate to the blockchain." When launched, the company's POLY token will also optimize for existing regulations, like KYC/AML.

Other teams working on security tokens include Overstock's tZero, CoinList, and Securitize. All said, security tokens appear to be less legally ambiguous than ICOs, especially amid regulatory inquiries. Top investors concur, and we expect more securitized token companies to emerge.

Proposed use cases for cryptocurrencies have yet to see material user traction. These range from decentralized payment networks to user-controlled social networks to prediction markets.

Many of the most profitable and well-funded blockchain companies have profited from speculation, but not use. These include exchanges, like Coinbase, or base-layer protocols, like Ethereum.

These companies made lots of money as a result of 2017's mania. Now, they're placing venture capital bets, going on shopping sprees, or building "ecosystem funds." All are trying to find use cases and users.

Exchanges are aware of rampant speculation, and are looking for real use cases via venture investment. Coinbase recently launched its venture arm, Coinbase Ventures. In early May, Huobi — a popular Chinese exchange — announced a \$1B fund to develop the Asian blockchain ecosystem. Binance — the largest exchange by volume — also announced an allocation of \$15M for the Bermuda blockchain ecosystem.

As bitcoin slides, exchanges acquire and invest



Base-layer protocols are still building working platforms. To incentivize dApp development, some are launching "ecosystem funds." Dfinity, which upon launch will compete with Ethereum, received \$61M in February from Andreessen and Polychain. Much of the money will go toward incentivizing developers to build on top of the Dfinity protocol. Similarly, Blockstack announced a \$25M fund last August to build out an app ecosystem on top of its platform. And, in early May, IOTA launched the IOTA Ecosystem to encourage development on its IoT-focused protocol.

in the past, which bodes well for Poloniex's future. Poloniex is one of the largest global exchanges, by trading volume.

Then, on April 13, Coinbase acquired dApp browser and cryptocurrency wallet Cipher. A few days later, the exchange acquired Earn (for a reported \$120M) and installed its CEO Balaji Srinivasan as Coinbase's new CTO. Earn has gone through multiple iterations. The company was first a bitcoin miner, then a chip manufacturer, and finally a social network where users got paid (in cryptocurrency) to answer emails. It's not yet clear how Coinbase will integrate Earn's line of business into its own. Still, the deal shows that Coinbase is gambling on sector use cases beyond speculation.

Coupled with regulatory considerations, we expect companies flush with capital will continue to invest in new and exciting use cases. Profitable companies that run regulatory risk will likely see bids, as evidenced by Circle's acquisition of Poloniex. Teams suddenly flush with cash will spend it on developing their respective ecosystems. All companies engorged on 2017's speculative frenzy will continue to deploy capital in search of blockchain's killer app.

Corporate activity

MENTIONS OF "BLOCKCHAIN" ON EARNINGS CALLS JUMP

As of late April, "blockchain" was already mentioned close to 300 times on Q1'18 earnings calls. Among other things, this indicates that corporates and analysts are, at the very least, talking about this technology with real interest.

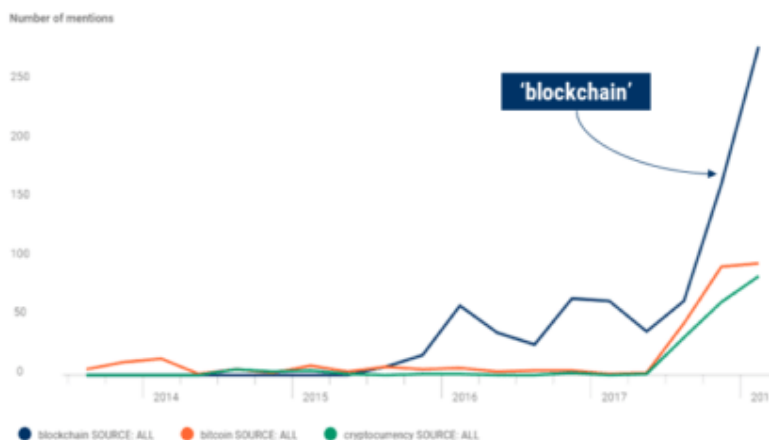
about blockchain

Besides for bitcoin, ICOs, and tokens, corporations were exploring use cases for blockchain technology in 2017 – and talking about it on earnings calls.

CBINSIGHTS

Mentions of 'blockchain' on earnings calls skyrocket

Q3'13 – Q1'18



Firms across verticals have referenced the technology on earnings calls, from Nasdaq to FedEx. Some companies even pivoted from their core products to adopt the technology, with (often suspect) stock price jumps to boot.

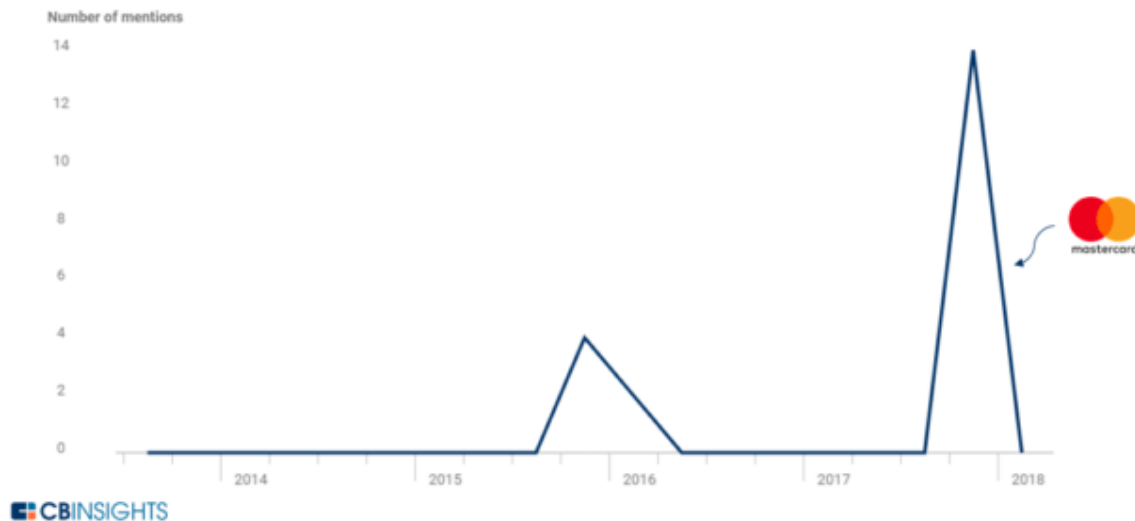
The recent spike, though, was unsurprising. Perhaps more notable is that a number of these companies were already talking about the technology in 2015, well ahead of last year's frenzy.

MasterCard CEO Ajay Banga highlighted the company's interest in the technology on a Q4'15 earnings call. "We've been experimenting in this space for a while. We've got patents in the space. We've got experiments and lab designs in the space. We've got investments with venture capital firms in the space."

Two years later, in Q4'17, Banga talked about blockchain at length. "We've got a bunch of things going on here: virtual cards, Mastercard Send, Vocalink, all of which are aimed at supporting cross-border and domestic B2B payment flows. And so that's what we're up to with the blockchain."

MasterCard follows blockchain hype cycle

Mentions of 'blockchain' on MasterCard's earnings calls. Q3'13 – Q1'18

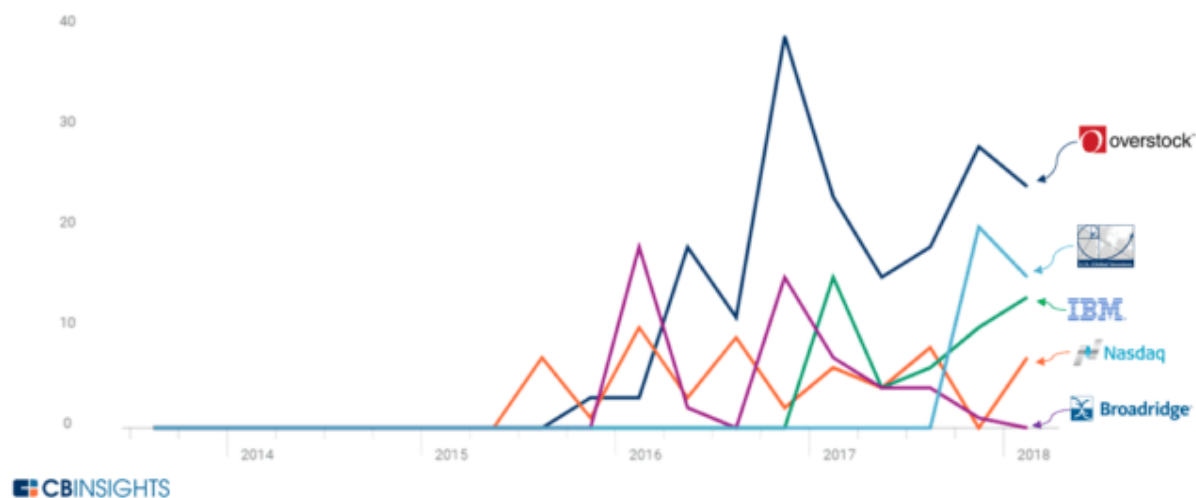


However, for the two years between these two earnings calls, MasterCard didn't mention blockchain technology once. Further, the company didn't mention blockchain technology in Q1'18 at all. This suggests that the firm's engagement with the technology hasn't matured, or rose and fell with broader market hype.

Other firms have exhibited a more sustained focus. Companies with the highest mentions of blockchain on earnings calls since 2013 are: Overstock (182 mentions), Nasdaq (57 mentions), Broadridge Financial Solutions (51 mentions), IBM (48 mentions), and US Global Investors (35). All — with the exception of US Global Investors — have talked about blockchain technology for a number of consecutive quarters.

Overstock talks the most about blockchain on earnings calls, but others are catching up

Mentions of 'blockchain' on earnings calls. Q3'13 – Q1'18



Overstock is going all in. The company's core line of business remains its e-commerce marketplace, but CEO Patrick Byrne is laser-focused on blockchain technology. Overstock announced that it would accept bitcoin payments in September 2014. In August 2017, it announced that it would accept other cryptocurrencies (like ethereum and litecoin). Accepting cryptocurrencies puts Overstock in a select group; cryptocurrencies still see limited transactional use, especially among top e-commerce merchants.

Byrne went so far as to describe Overstock as a "holding company" on the company's Q4'16 earnings call. "Think of [Overstock] as a holding company with on one side an [...] internet retail company that is spinning off \$75 million a year. That's on one side. On the other side is Medici Ventures, which has a number of investments in blockchain and fintech."

Another company that's talked a lot about blockchain technology is Nasdaq. The company tested the technology in its Nasdaq Private Markets (NPM) product, but earnings call transcripts and other data points suggest lagging deployment.

In Q2'15, the stock exchange said that it was "exploring the use of blockchain technology in Nasdaq Private Markets [and] targeting initial release of this technology in the fourth quarter of 2015." Mentions of NPM (Nasdaq Private Markets) and blockchain went hand in hand for a number of quarters as the technology was beta-tested.

By Q1'16, the technology was labeled a success in that quarter's earnings call. "We continue to enhance this platform in new and innovative ways to reduce the administrative complexities and costs private companies face today. This success follows last quarter's successful execution of the first blockchain-enabled transaction in the private space."

2016, the NFF helps clients integrate business processes, like clearance and settlement, with new technologies. NFF is part of a broader Nasdaq shift toward fintech. By 2017 in Nasdaq's 10-K, blockchain technology was not mentioned at all in relation to NPM. All this perhaps indicates limited traction and deployment, or a complete strategic shift toward its NFF product.

In the same breath, Nasdaq is taking a more considered view of cryptocurrencies, and cryptocurrency trading. As Nasdaq president and CEO Adena Friedman said in an April interview: "I believe that digital currencies will continue to persist...it's just a matter of how long it will take for that space to mature [...] once you look at it and say, 'do we want to provide a regulated market for this?' Certainly, Nasdaq would consider it."

With all this, talk is still cheap. It remains to be seen what's corporate theater, and what's not. One measure of real corporate interest might be investment activity, which we'll dive into in our next trend.

CORPORATE INVESTMENTS AND PARTNERSHIPS SKYROCKET – BUT WE'VE BEEN HERE BEFORE

119 corporations and corporate venture groups invested in blockchain technology in 2017 – the highest annual total yet. Even further, Q1'18 saw a little less than half that, implying that 2018 is going to be a huge year of corporate dealmaking in the sector.

Total corporate investors jumped 70% YoY between 2016 and 2017

Banks and other financial institutions participated in big rounds to R3 in 2015 and 2017 (\$107M Series A), pushing the total number of corporate investors up.

CBINSIGHTS



This recent spike might imply that corporates are new entrants to the conversation, but many have been around for quite some time. Long enough in fact that some early – and highly-touted –

In January 2017, Du Telecom partnered with NMC Healthcare to explore blockchain-based health records. Yet, the former project lead at Du has since left the firm, and there have been no further announcements regarding the project. The project was slated to launch in early 2018.

Du Telecom partnered on blockchain for health, but found limited success

In January 2017, Du announced a partnership with NMC Healthcare to implement electronic health records using blockchain. Estonian blockchain technology startup Guardtime would also play a part in the project, which was slated to launch in Q1'18. However, the former project lead at Du, has since left the firm, and there have been no further announcements regarding the project.



"By digitizing all health records, and putting them in blockchain technology, data can be shared and distributed across all hospitals. **We are looking at fully digitizing the health system with blockchain technology to assure data integrity with zero error and guarantee end-to-end accountability of the patients' records.**"



Carlos Domingo
(Former) Chief New Business and Innovation Officer at Du



A similar story played out at mining company BHP Billiton. BHP started its blockchain project in 2016 to enhance security and tracking around real-time mining data. In April 2017, BHP's project lead left to join Consensusys (an Ethereum development firm), and no further news has been published about the project.

Heavy industry is exploring blockchain but some projects appear to have stalled

At the end of 2016, mining firm BHP Billiton partnered with BlockApps and Consensys to apply blockchain to its supply chain and enhance security around real-time mining data. BHP hoped to coordinate disparate contractors on a blockchain-based system, but in April 2017 the blockchain head at BHP left to join Consensys (an Ethereum development firm) as director of energy and consulting.



Financial services firms have also been challenged. Rumor has it that JPMorgan is mulling a spinoff of its in-house blockchain project, Quorum. Amber Baldet, who headed the project, left the firm in early April. One issue cited by the Financial Times is that banks are wary of using a distributed ledger run by a rival bank for mission-critical data.

Other corporates are looking beyond consortia and enterprise use cases. In Q4'17, Theta Labs received \$8M in seed money from the Sony Innovation Fund and Bertelsmann Digital Media Investments (among others). Theta Labs is building a decentralized video streaming platform, and using a native token (THETA). This was the first blockchain technology investment for both these investors.

As more enterprise players invest in a new batch of corporate initiatives, we expect that many will find familiar problems. A running theme throughout this report is that blockchain technology isn't new — but its increasing prominence is. As a result of big price increases, more stakeholders have come to the table. The question remains: what is blockchain technology's killer app? Corporates should think long and hard before deploying what might be a solution in search of a problem.

WHILE CONSORTIA FORGE AHEAD, QUESTIONS REMAIN

Corporates are once again investing in consortia. As highlighted above, 119 corporates deployed capital to blockchain companies in 2017. Of those, 39 participated in a \$107M Series A to the financial services software provider and consortium R3. Of note, a similar story played out in 2015: 89 corporates placed direct equity investments in the sector, with 42 investing in R3.

At the DTCC's Fintech Symposium, co-founder of R3 Todd McDonald spoke about some of the legal issues some pilots have faced, even as the technology itself worked as hoped. "From the legal side, there were about seven or nine pieces of paper that had to get signed for [...] one transaction. None of this is easy. We need to bring in the existing legal constructs. It's still a lot of work that has to be done."

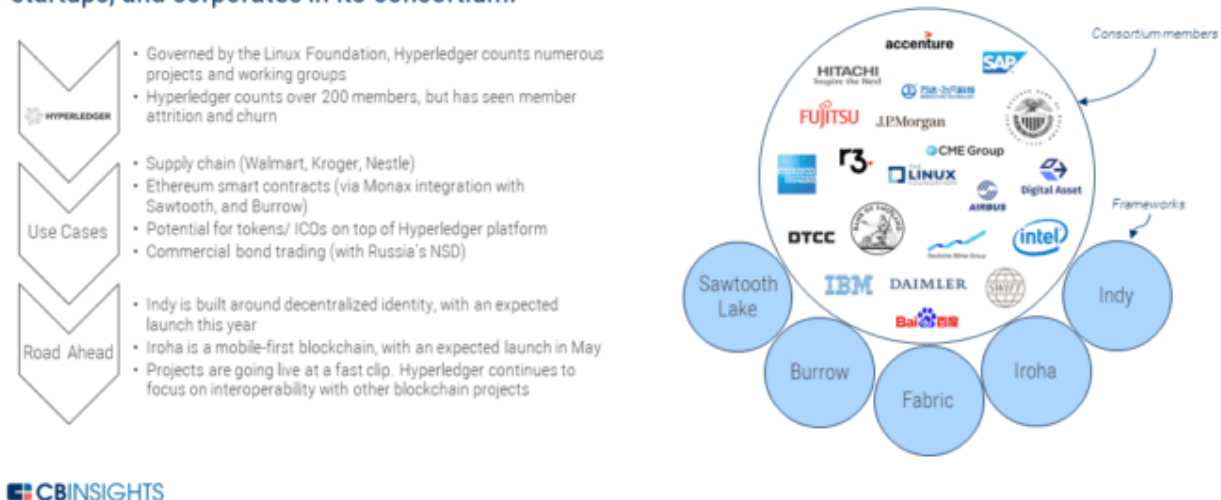
Finastra), as well as for cross-border payments.

Another consortium, the Enterprise Ethereum Alliance boasts over 500 members. Although launched more than a year ago, the consortium has been quiet, and only just announced its first “roadmap” that outlines its open standards work.

Governed by the Linux Foundation, Hyperledger is an open-source, cross-industry effort to create distributed ledger frameworks for enterprises. The consortium now counts 200+ members and various working groups exploring distributed ledger use cases. While Hyperledger saw some members churn in 2017, it also added 60 members and 3 new frameworks: Burrow, Indy, and Quilt.

Hyperledger finds first use case in supply chain

Hyperledger’s Fabric framework is geared toward supply chain. The consortium has seen members come and go – by the end of 2017, 15+ members had left or downgraded membership. Still, Hyperledger is adding new members, and now counts over 200 banks, startups, and corporates in its consortium.



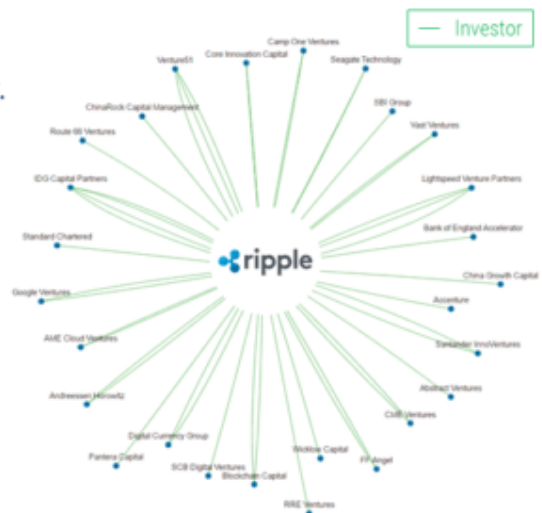
Some IT consultancies use Hyperledger in their own blockchain offerings. IBM Blockchain is a commercial version of Hyperledger. Last year, Walmart, Kroger, and Nestle used IBM Blockchain to track food through the supply chain. Most recently, Chinese tech giant Huawei announced a Hyperledger-powered blockchain. This will allow clients to build smart contract applications around supply chain, among other use cases.

Ripple is an outlier in this cohort; it's not a consortium, but assembles consortia which use its software. Ripple has received \$94M in venture financing to build a cross-border payments solution. In March, Ripple said that it was building a distributed payments app for speedy transactions between a group of 60+ Japanese banks.

Ripple partners on pilots, but finds some critics

Ripple has received \$94M in venture financing since 2013 to build cross-border payments software, and has assembled consortia and partnerships using its technology.

Controversially, **the company also owns a majority of its associated cryptocurrency, XRP**, which peaked in January at a \$137B market capitalization – well above Ripple's \$255M private market valuation.



CBINSIGHTS

Ripple's cryptocurrency, XRP, has caused some controversy. While Ripple has a private market valuation of around \$255M (as of its \$55M Series B in September 2016), XRP has a market capitalization of \$26B (as of 4/17/18). The market is valuing Ripple's software at a much lower price than its associated cryptocurrency. The vast difference between the two valuations demonstrates a broader market shift, as investors now focus on cryptocurrencies (like XRP), not blockchain software (like Ripple).

Ripple owns a majority of outstanding XRP and has deployed it in unexpected ways. In one instance, the company invested \$25M in home storage and rental startup Omni. In another, it invested \$25M in Blockchain Capital, one of the company's own investors.

With all this, enterprise-focused consortia are facing an uphill battle.

Some corporates have concluded that blockchain technology might be a solution looking for a problem. Citing cheaper alternatives, the DTCC recently shelved a blockchain for repurchase agreement transactions. Last year, a partnership between blockchain provider Paxos and Euroclear fell apart.

For R3, the EEA, Hyperledger, and Ripple, 2017 was a big year – but the question remains: which of these consortia will be the first to ship mass-use products?

In the meantime, as consortia bolster their ranks and explore use cases, they should also focus on interoperability. For a technology built around cooperation, it most likely won't be a winner-take-all scenario.

The road ahead

building new use cases, and corporates are — once again — approaching the drawing table.

Investors should remain cautious, though. Most problems likely don't require blockchain-based solutions. That being said, 2017's cash infusion should provide runway as teams look for blockchain's killer app.

YOU MIGHT ALSO LIKE:

[Industrial IoT & The Factory Of The Future](#)

[Corporate Blockchain Investment Is Up Again, Though First-Movers Face Growing Pains](#)

[Global Fintech Report Q1 2018](#)

COMMENTS

0 comments

WHERE IS THIS DATA COMING FROM?

Start your free trial today

Business Email

SIGN UP

Research Portal

cbinsights.com

